CAEA: Auditing and Repairing Preference-Induced Belief Feedback

Xiaotong Zhang
China University of Petroleum (East China)
Qingdao, China

Abstract

Pooled one-turn audits can hide feedback instability. We construct policies with identical pooled response–outcome laws but different mean-response stability, including an arbitrarily-small-regret extension. Conflict-aware epistemic alignment (CAEA) repairs an externally audited content marginal while retaining within-class policy. Classical KL projection accounts for preference cost; coupling bounds relate repair to contraction, and a finite audit certificate requires supplied curvature. Controlled DPO and stochastic simulations are complemented by 4,672 unique fresh scoring requests on three frozen 1.5B–7B checkpoints, with 256/64/128 Bayesian training/validation/test cases and 32 external correctness probes. A separately calibrated auditor uses no test labels. Constant-target, answer-order, informative-evidence and auditor-refit controls distinguish target quality from algebraic repair identities. The learned audit does not establish an advantage over simple shrinkage and strongly attenuates informative updates. The contribution is an audit-to-feedback analysis and reproducible failure study, without claims of full-model post-training or human benefit.

1 Introduction

An assistant can give acceptable answers in isolation yet reinforce a user's unsupported belief over repeated exchanges. Human preference judgments can reward agreement at the expense of truthfulness (Sharma et al., 2024), creating a plausible mechanism for such feedback. The mechanism is conditional: agreement can be warranted, and preference optimization can also reward evidence-based correction. The relevant question is therefore when preference adaptation changes the stability of an interaction, rather than whether preference alignment and epistemic quality are inherently incompatible.

Consider users who receive the same evidence but express different beliefs. If an assistant responds to each user's unsupported stance, averaging its answers across users may conceal systematic differences in what each user receives. We formalize this distinction with a binary construction: the pooled one-turn response–outcome distribution, including its Brier risk, remains unchanged as agreement pressure increases, while the mean-response interaction acquires separated stable equilibria. A pooled audit and a feedback analysis can therefore give different answers about the same policy.

We study a corresponding intervention, conflict-aware epistemic alignment (CAEA). An external audit partitions responses into epistemic classes and specifies a target distribution over those classes. CAEA changes the class marginal while retaining the preference policy conditional on each class. This permits a precise distinction between changing an answer's audited content and changing its remaining attributes. In the experiments, classes encode probability reports and within-class variation encodes abstract style; whether this partition captures all information affecting a human reader is a separate empirical question.

The analysis connects three quantities: the epistemic marginal being enforced, the preference cost of changing it, and the intervention strength needed for a specified contraction margin. It also identifies what an audit must assume. Finite observations at selected stances cannot certify global stability for an unrestricted smooth response function; a supplied curvature bound makes a conservative finite-sample certificate possible.

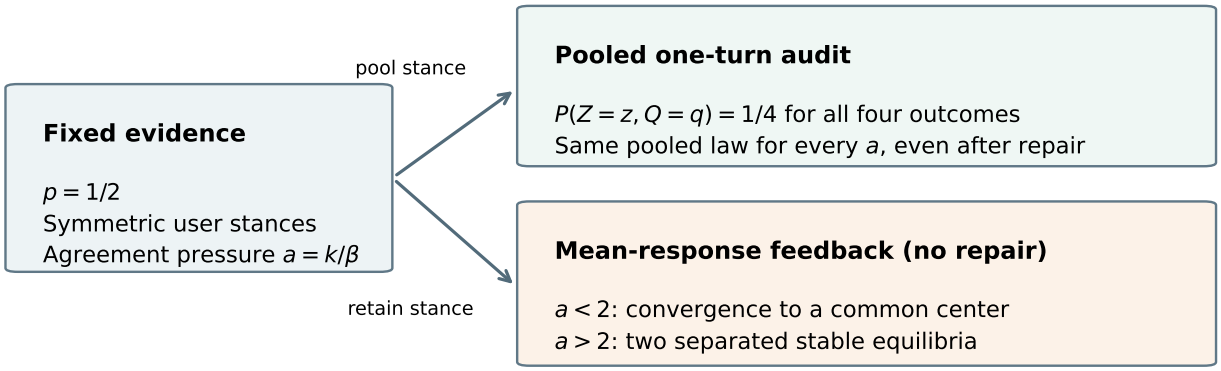


Figure 1: Two views of the same controlled binary model. Pooling over a symmetric stance population hides agreement pressure; retaining stance reveals different mean-response feedback regimes. The displayed threshold uses zero repair. This is a model counterexample, not a claim about human belief thresholds.

Contributions. The central result is a counterexample to identifying feedback stability from a pooled one-turn joint distribution, not a claim that preference optimization is universally harmful. We connect this observability gap to (i) an exact repair threshold in the binary mean-response model, (ii) preference-cost accounting and contraction bounds for conditional repair, and (iii) a finite-audit impossibility result and a certificate conditional on supplied regularity. The KL projection itself is classical, and closest-KL anti-sycophancy correction is prior work. The coupling and finite-grid arguments use standard proof techniques; we claim their explicit audit-to-feedback connection rather than a new general projection or contraction principle. Our analysis asks what the audit observes, what intervention its target requires, and which feedback guarantee follows. Controlled training and simulations test the implementation. A larger fresh-inference study separately calibrates a common auditor, evaluates three checkpoints up to 7.61B parameters, and retains paired stance, external-task and adverse-baseline controls.

2 Related work

Preference optimization and sycophancy. DPO expresses preference learning through a policy/reference likelihood ratio (Rafailov et al., 2023). Empirical evidence links sycophancy to human preference judgments (Sharma et al., 2024). Shapira et al. analyze behavioral amplification and derive a closest-KL correction that prevents increased agreement (Shapira et al., 2026); their result is the closest antecedent to our projection component. Our emphasis is the connection between a class-marginal repair and longitudinal feedback, including the observability limits of pooled and finite stance audits. Moya et al. identify spurious-learning mechanisms and a tie-training intervention (Moya et al., 2026), highlighting that confirmation-biased labels are not the only source of undesirable drift.

Belief updating and multiple perspectives. BASIL distinguishes sycophantic shifts from Bayesian updating and studies calibration and post-training (Atwell et al., 2026). Ma et al. separate unsupported yielding from rational updating (Ma et al., 2026); PlurPO constructs preferences using multiple stakeholder perspectives (Hatgis-Kessell et al., 2026). These approaches motivate evaluating selectivity rather than suppressing agreement indiscriminately. Our evidence-fixed intervention changes a user's unsupported stated stance, while informative user input belongs in the evidence. Ghosh et al. investigate nonlinear belief dynamics under sycophantic feedback (Ghosh et al., 2026). We focus on a response-policy construction with explicit audit equivalence and repair-budget calculations. Their attractor analysis already addresses nonlinear sycophantic feedback; the occurrence of separated stable beliefs alone is therefore not our novelty claim.

Projection, scoring, and diversity. The repair follows from the KL chain rule and information projection (Csiszár, 1975). Forecast evaluation uses proper-score regret (Gneiting & Raftery, 2007). Output-diversity

methods such as DivPO (Lanchantin et al., 2025) address a different property: a broad distribution of answers need not be well calibrated. We use standard statistical quantities to keep these distinctions explicit. Contraction of random iterations is classical (Diaconis & Freedman, 1999); the coupling below specializes that reasoning to a shared audit target.

Broader evaluations. SYCON evaluates multi-turn free-form sycophancy (Hong et al., 2025); our finite answer banks do not substitute for that setting. External probes below are a transformed GSM8K subset (Cobbe et al., 2021), not standard solve accuracy.

3 Problem formulation

Let x denote shared admissible evidence and $s \in [0,1]$ a user's stated probability. Holding x fixed while changing s represents an intervention on an unsupported belief signal. Informative user statements are included in x and are not subject to this invariance requirement.

The response space $\mathcal{Y}$ is finite. An audited map $c_x : \mathcal{Y} \to \{1, \dots, C\}$ assigns each response to an epistemic class. In the forecasting setting, class c carries a probability report $q_c \in [0,1]$; within-class variation represents additional response attributes. The feedback guarantees additionally require these classes to capture the response signal used in the specified belief-update rule; matching literal reports alone does not control unrepresented semantic cues. Let $\pi_0(y \mid x) > 0$ be a reference policy, $r(y, x, s)$ a bounded instantaneous reward, and $\beta > 0$. Define

$$J_{x,s}(\pi) = \mathbb{E}_\pi r(Y, x, s) - \beta D_{\mathrm{KL}}(\pi \| \pi_0), \qquad \pi_P(y \mid x, s) = \frac{\pi_0(y \mid x) e^{r(y,x,s)/\beta}}{Z_{x,s}}. \tag{1}$$

The Gibbs policy π_P is the ideal distributional maximizer. Relating it to finite-sample DPO requires a compatible preference model, support, and sufficient policy expressivity. Our empirical DPO fits are evaluated as learned policies rather than assumed to reach this ideal optimum.

Write $m_P(c \mid x, s) = \sum_{y : c_x(y) = c} \pi_P(y \mid x, s)$. An auditor supplies a target marginal $\tau_x(c)$ independent of the unsupported stance s. The auditor and the class map are inputs to CAEA. In the synthetic oracle experiments, the Bernoulli evidence probability p_x is known, and the target is

$$\tau_x(c) \propto \exp\{-(q_c - p_x)^2 / T_E\}, \qquad T_E > 0. \tag{2}$$

Here T_E is a chosen exploration tolerance. Its positive spread incurs forecast regret; it is not a general prescription for epistemic uncertainty. Strong evidence can justify a concentrated target. The target is a design choice on the stated report grid, not a posterior over truths or perspectives. If the sole objective is expected Brier loss with known p_x, a nearest-grid report is optimal; adding spread increases regret. Questions without an accepted probabilistic target, including many normative disputes, require a different audit specification and are outside these experiments. The expanded study substitutes an independently calibrated estimate $\widehat{p}_x$ for p_x; test truth evaluates non-oracle methods but never constructs their targets.

4 Conditional epistemic repair

For a repair fraction $\lambda \in [0,1]$, set

$$b_\lambda = (1 - \lambda) m_P + \lambda \tau_x, \qquad \pi_\lambda(y \mid x, s) = b_\lambda(c_x(y) \mid x, s) \pi_P(y \mid c_x(y), x, s). \tag{3}$$

Full repair ($\lambda = 1$) replaces the epistemic marginal with the stance-independent target. Partial repair interpolates between this target and the preference marginal. Both preserve every positive-mass class-conditional response distribution exactly.

Identities, not independent predictive evidence. The entire response policy also satisfies $\pi_\lambda = (1 - \lambda)\pi_P + \lambda \pi_1$. For fixed evaluation weights over (x, s), linearity and homogeneity give

$$\mathcal{R}(\pi_\lambda) = (1 - \lambda)\mathcal{R}(\pi_P) + \lambda \mathcal{R}(\pi_1), \tag{4}$$

$$\mathrm{TV}(b_\lambda, \tau_x) = (1 - \lambda)\, \mathrm{TV}(m_P, \tau_x). \tag{5}$$

Thus, intermediate risk and target-TV improvements follow algebraically once the target is supplied. Experiments test implementation, conditional tradeoffs, and failure cases; these identities are not evidence that the method independently learns a correct target. Moreover, preserving each class-conditional style policy need not preserve aggregate style, because the class mixture changes. Equation (4) also implies $\mathcal{R}(\pi_\lambda) - \mathcal{R}(\pi_P) = \lambda[\mathcal{R}(\pi_1) - \mathcal{R}(\pi_P)]$. A less accurate target therefore increases report regret for every positive repair fraction, even when it removes unsupported stance dependence.

Proposition 1 (KL projection and preference cost)**.** *For fixed (x, s) and a target marginal b supported by m_P, the policy $\pi_b(y) = b(c_x(y))\pi_P(y \mid c_x(y))$ is the unique minimizer of $D_{\mathrm{KL}}(\pi \| \pi_P)$ among policies with class marginal b. Moreover,*

$$J(\pi_P) - J(\pi_b) = \beta D_{\mathrm{KL}}(b \| m_P). \tag{6}$$

Full target matching and preference optimality coincide exactly when $m_P = \tau_x$.

The proposition instantiates classical information projection (Csiszár, 1975). Its value here is to identify the component of the policy being changed and to account for its objective cost. Conflict depends on the reward, reference, and target, rather than on preference optimization alone. Full proofs appear in Appendix A.

Response selection. After scoring a bank of N candidates, normalize the preference scores within each class, sample a class from b_λ, then sample a response from its existing class-conditional distribution. The additional computation is $O(N)$. Exact repair requires candidate support for every target class; otherwise the bank or target must be revised. Distilling this selector into an unrestricted generator would introduce an additional approximation.

5 Feedback hidden by pooled evaluation

We first isolate the distinction between pooled output quality and stance-conditioned feedback in a binary response model. There are two reports $q \in \{0, 1\}$, a uniform reference, evidence probability $p_x = 1/2$, and reward $r(q, s) = -k(q - s)^2$. Let $a = k/\beta$. Then

$$m_a(1 \mid s) = \mathrm{sigmoid}(a(2s - 1)). \tag{7}$$

Under full repair, $\tau(1) = 1/2$; under partial repair, the expected report is $f_\lambda(s) = (1 - \lambda)m_a(1 \mid s) + \lambda/2$.

Theorem 2 (Pooled audit blindness and a repair threshold)**.** ***Pooled observations.*** *Let the population stance S have a distribution symmetric about $1/2$. Conditional on the fixed evidence, let $Z \sim \mathrm{Bernoulli}(1/2)$ be jointly independent of the stance and response-sampling randomness. For every $a \geq 0$ and every λ, the pooled joint distribution of (Z, q) is the same: both coordinates are independent uniform binary variables. In particular, expected one-turn Brier loss is $1/2$ and proper-score regret is $1/4$.*

Mean-response feedback. *For the deterministic update*

$$s_{t+1} = F_\lambda(s_t) = (1 - \eta)s_t + \eta f_\lambda(s_t), \quad 0 < \eta \leq 1, \tag{8}$$

the derivative at $s = 1/2$ is $1 - \eta + \eta(1 - \lambda)a/2$. If $(1 - \lambda)a < 2$, F_λ is a strict global contraction. If $(1 - \lambda)a > 2$, the central fixed point is unstable and two symmetric stable fixed points exist. Thus, the infimum repair fraction for strict contraction along this mixture family is $\lambda_{\mathrm{crit}} = \max\{0, 1 - 2/a\}$ for $a > 0$; equality at $(1 - \lambda)a = 2$ does not certify a geometric rate. For $a = 0$, no repair is needed for stability.

Corollary 3 (The gap can persist near optimal forecast risk)**.** *Fix $p \in (0, 1)$ and $0 < d \leq \min\{p, 1 - p\}$. Restrict reports to $q \in \{p - d, p + d\}$ and stances to $[p - d, p + d]$, with a uniform reference, reward $-k(q - s)^2$, and symmetric stance population about p. Set $c = 2kd^2/\beta$. Pooling gives each report probability $1/2$ for every k and every repair fraction, independent of $Z \sim \mathrm{Bernoulli}(p)$. Pooled Brier loss is $p(1 - p) + d^2$ and regret is d^2. For the same mean-response update and uniform audit target, strict contraction holds if $(1 - \lambda)c < 1$, while the center is unstable and two stable fixed points exist if $(1 - \lambda)c > 1$.*

The report regret can therefore be arbitrarily small while the normalized feedback regimes differ. Maintaining a fixed unstable c as $d \to 0$ requires k/β to grow as d^{-2}; absolute equilibrium separation is at most $2d$. This extension does not establish large real-world effects from small forecast errors at fixed agreement pressure.

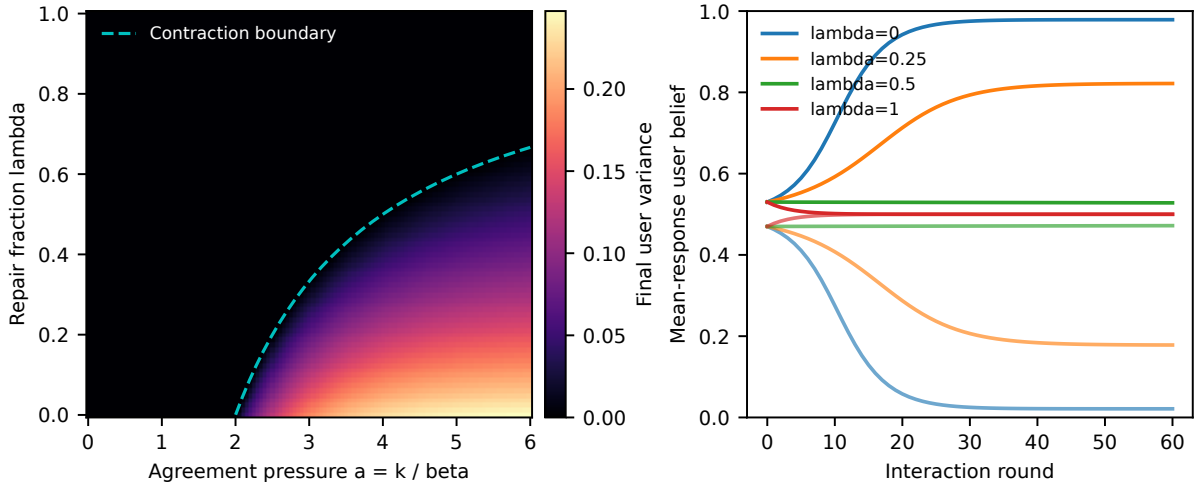


Figure 2: Deterministic mean-response dynamics. Left: final population variance after 400 updates from $s_0 \in \{0.47, 0.53\}$, $\eta = 0.25$. The dashed line is the strict-contraction boundary. Finite-horizon variance remains nonzero near the boundary. Right: paired trajectories at $a = 4$. The simulations use the specified belief-update rule.

Thus, tests based only on the pooled one-turn joint law cannot distinguish the constructed feedback regimes. Conditioning on stance or retaining linked longitudinal observations restores information that pooling discards. The threshold characterizes this mean-response model; it is not an estimated threshold for human interaction. The assistant policy is fixed throughout the interaction rather than retrained after each round.

6 Stability guarantees and audit requirements

Let $K_{x,s}$ be the preference policy's distribution over report values q_c. In kernel formulas, τ_x denotes its report pushforward $\sum_c \tau_x(c)\delta_{q_c}$, not a distribution on response text. Assume a finite L_x satisfies $W_1(K_{x,u}, K_{x,v}) \le L_x|u - v|$, where W_1 is Wasserstein distance on $[0, 1]$. The same λ_x must be used across non-evidential stances. A stance-dependent repair fraction would introduce an additional feedback term and requires a separate analysis.

Theorem 4 (Stability budget and approximate full repair)**.** *Fix $0 < \eta \le 1$. For the sampled update $U' = (1-\eta)U + \eta Q$, the Markov kernel induced by $K^{\lambda}_{x,s} = (1-\lambda_x)K_{x,s} + \lambda_x\tau_x$ is W_1-Lipschitz with coefficient at most*

$$\rho_x = 1 - \eta + \eta(1 - \lambda_x)L_x. \tag{9}$$

For a common evidence sequence with $\sup_x \rho_x < 1$, differences between initial belief distributions contract geometrically in W_1. This compares belief laws, not the variance of independently sampled individuals.

Under full repair, suppose the implemented target $\widehat{\tau}_{x,s}$ differs from a stance-independent ideal target by at most ϵ_t in total variation at round t, uniformly over current belief states. Starting with the same initial belief and common evidence, its belief law and the ideal target-driven law can be coupled so that

$$\mathbb{E}|U_T - \widetilde{U}_T| \le \eta \sum_{t=0}^{T-1} (1-\eta)^{T-1-t}\epsilon_t. \tag{10}$$

With a constant error bound, this is at most $\epsilon[1 - (1-\eta)^T]$.

If L_x is known, a desired margin $\gamma \in (0, 1)$ is obtained by choosing $\lambda_x \ge \max\{0, 1 - (1-\gamma)/L_x\}$ for $L_x > 0$, giving $\rho_x \le 1 - \eta\gamma$. By convexity, $D_{\mathrm{KL}}(b_\lambda \| m_P)$ is nondecreasing along $\lambda \in [0, 1]$. Consequently, the smallest admissible fraction minimizes the exact cost *along this mixture path*; no optimality over all stable policies is asserted. A positive margin avoids the excluded boundary of a strict-contraction condition.

Corollary 5 (A finite-audit expectation bound)**.** *For $n \geq 1$ independent class samples from a trusted target τ over C classes, the empirical frequency $\bar{\tau}$ satisfies $\mathbb{E}\,\mathrm{TV}(\bar{\tau}, \tau) \leq \frac{1}{2}\sqrt{(C-1)/n}$. For the add-$\alpha$ estimator $\widehat{\tau} = (n\bar{\tau} + \alpha\mathbf{1})/(n + C\alpha)$ with $\alpha \geq 0$, an upper bound is*

$$\mathbb{E}\,\mathrm{TV}(\widehat{\tau}, \tau) \leq \min\left\{1, \frac{1}{2}\sqrt{\frac{C-1}{n}} + \frac{C\alpha}{n + C\alpha}\right\}. \tag{11}$$

This controls target estimation error in expectation, not auditor bias.

Stability and correctness are distinct. An incorrect but stance-independent target can make the system stable and wrong. Also, if unmeasured wording changes the interpreted epistemic signal, the report-class map is not sufficient and Theorem 4 does not cover that mechanism. Auditor validity and the sufficiency of the class map are therefore separate requirements.

6.1 From finite response audits to a stability certificate

Equation (9) requires a global sensitivity bound L_x. Finite stance tests alone cannot establish it, even if the response probability at every tested stance is known exactly.

Proposition 6 (Finite stance audits need regularity)**.** *For any finite set of audited stances $A \subset [0,1]$, there is a smooth, bounded binary response probability $m(s)$ with $m(s) = 1/2$ at every $s \in A$, yet with two stable off-grid fixed points for $F(s) = (1-\eta)s + \eta m(s)$, for every $0 < \eta \leq 1$. Their separation can be arbitrarily close to one. Thus, exact agreement with an invariant target on finitely many stances cannot by itself certify global feedback stability.*

Conversely, let $m \in C^2([0,1])$ satisfy a known global curvature bound $|m''(s)| \leq H$. Audit a uniform grid $s_j = (j-1)/(J-1)$ of $J \geq 2$ points, spacing $h = 1/(J-1)$, with n independent binary responses per point. Let $\widehat{m}_j$ be the empirical means and $\epsilon = \sqrt{\log(2J/\delta)/(2n)}$ for $0 < \delta < 1$. With probability at least $1-\delta$,

$$\sup_s |m'(s)| \leq \widehat{L}_H := \max_j \frac{|\widehat{m}_{j+1} - \widehat{m}_j|}{h} + \frac{2\epsilon}{h} + Hh/2. \tag{12}$$

This can be inserted into Equation (9) for binary reports, where the kernel W_1 distance equals the difference of response probabilities.

The $2\epsilon/h$ term penalizes overly fine grids at fixed observations per point; $Hh/2$ penalizes overly coarse grids. For total report budget $N \simeq Jn$, balancing the two errors gives h of order $N^{-1/5}$ up to logarithms when $H > 0$ is fixed. A trustworthy curvature bound is additional information: estimating it from the same unconstrained finite grid does not evade the first part of the proposition. The certificate consequently trades sampling precision against coverage under a supplied regularity assumption. It is pointwise in evidence context x; simultaneous certification of M prespecified contexts requires, for example, a union bound using δ/M. It does not certify an unrestricted set of unseen contexts.

7 Measuring report quality and belief dispersion

We use three complementary quantities based on standard variance, total-variation, and proper-scoring constructions. They describe different objects: simulated belief dispersion, audited class coverage, and forecast error. The legacy BPD/PCD/ECR field names are retained for data compatibility, not introduced as validated cognitive metrics. We use ordinary dispersion, target mismatch, and regret in the interpretation.

Excess belief dispersion (legacy output field: BPD). For a common evidence context and matched initial user population, let $V_T^\pi(x) = \mathrm{Var}(U_T^\pi \mid x)$ and let $\widetilde{\pi}$ use the trusted, stance-independent target with the same update rule and sampling convention. Define

$$\mathrm{BPD}_T = \mathbb{E}_x[V_T^\pi(x) - V_T^{\widetilde{\pi}}(x)]. \tag{13}$$

BPD is signed and measures excess conditional dispersion. Moving all users toward the same wrong answer need not increase BPD, so user mean-squared evidence error must also be reported. For stochastic updates, subtract the oracle's nonzero sampling variance; raw change from initial variance is also retained in the released results.

Conditional target mismatch (legacy difference field: PCD). Define $d_\pi(x,s) = \mathrm{TV}(m_\pi(\cdot \mid x,s), \tau_x)$ and $\mathrm{PCD} = \mathbb{E}[d_\pi - d_{\pi_0}]$. Also report absolute $\mathbb{E}[d_\pi]$. Here "perspective coverage" is operationalized only as forecast-class coverage relative to the evidence target; semantic viewpoint coverage would require an independently validated class map. Uniformly spreading mass across unsupported answers can worsen this quantity. Conditioning on (x,s) is essential; pooling first can erase the signal in Theorem 2.

Brier regret and change from reference (legacy field: ECR). For $Z \mid x \sim \mathrm{Bernoulli}(p_x)$ and a report sampled independently of the realized Z conditional on x,

$$\mathcal{R}(\pi) = \mathbb{E}_{x,s,Y\sim\pi}(q_{c_x(Y)} - p_x)^2, \quad \mathrm{ECR} = \mathcal{R}(\pi) - \mathcal{R}(\pi_0). \tag{14}$$

This equals excess expected Brier loss over the evidence posterior, followed by subtraction of the reference regret. It is not identical to the reliability component of an aggregate calibration decomposition and does not measure hedging frequency. In the binary blindness example, ECR is zero even while BPD grows; report and user measurements answer different questions.

8 Experiments

8.1 Finite-action preference learning

There are 21 forecast classes $q \in \{0, 0.05, \ldots, 1\}$ and three abstract styles $z \in \{-1, 0, 1\}$, giving 63 responses. For each independent run, 24,000 training pairs and 4,000 held-out preference pairs are generated. Each training context samples $p = 0.05 + 0.9B$, $B \sim \mathrm{Beta}(0.8, 0.8)$, and $s \sim \mathrm{Uniform}(0.05, 0.95)$. The reference has

$$\pi_0(q, z \mid p) \propto \nu(z) \exp[-(q - p_{\mathrm{ref}})^2/0.06], \quad p_{\mathrm{ref}} = 0.5 + 0.65(p - 0.5), \quad \nu = (0.25, 0.5, 0.25). \tag{15}$$

The reward is

$$r = -(q-p)^2 - k(q-s)^2 + 0.8(2s-1)z + 0.25(2q-1)(2s-1)z, \tag{16}$$

with $k = 4$ and $\beta = 0.5$ in the main setting. Pair labels follow Bradley–Terry probabilities $\mathrm{sigmoid}(r(y_a) - r(y_b))$. Candidate proposals are a 50:50 mixture of reference and uniform response sampling. The six supplied features contain the simulator reward exactly, up to a context-only constant: the ideal coefficient vector is $(0, -(1+k), 2, 2k, 0.8, 0.25)/\beta$. This is a well-specified low-dimensional learning test with direct access to p, not a test of learning evidence from text. We fit the DPO pairwise likelihood using L-BFGS-B in the log-linear policy $\pi_\theta \propto \pi_0 \exp(\theta^\top \phi)$ with $\phi = (q, q^2, qp, qs, z(2s-1), z(2s-1)(2q-1))$. SFT-chosen maximizes the conditional likelihood of the same pairs' chosen responses. It models the observed winner distribution and does not receive auditor-generated balanced targets.

Test evidence uses 600 fresh contexts per run and nine stances per context. Ten seeds yield 54,000 context–stance evaluations. The auditor target uses Equation (2) with $T_E = 0.01$. Audit-only uses this same target and the reference style prior; CAEA uses the learned DPO class-conditionals. Noisy32 estimates p from 32 Bernoulli outcomes using a $\mathrm{Beta}(1/2, 1/2)$ posterior mean, shared across counterfactual stances. Audit-only-noisy32 uses the identical noisy target and reference style prior, equalizing target information with CAEA-noisy32. Oracle-nearest uses the known p to select a closest grid report and retains the DPO conditional; it optimizes report risk rather than the prescribed spread target. Pooled-marginal repair geometrically averages the DPO class marginals over the nine stances and retains DPO within-class distributions. Uniform-coverage sets all forecast classes equally likely. Soft-score baselines reweight DPO by $\exp[-v(q-p)^2/\beta]$ for $v \in \{0.5, 2, 8\}$; all settings are reported, with no post-test selection of a winner.

For fitted DPO policies, reported KL is policy displacement. The exact objective-cost identity applies to the fitted policy's implicit reward; it need not equal a gap under the simulator's true reward after finite-data estimation error.

Table 1: Finite-action results: means ± 95% Student-t confidence half-widths over ten independent runs. Style is only the style component of Equation (16), not measured human helpfulness. Target information is given to audit-based methods. KL is relative to learned DPO.

Method	Proper-score regret ↓	Style reward ↑	Target TV ↓	KL to DPO
Reference	0.0388 ± 0.0001	0.0000 ± 0.0000	0.4731 ± 0.0008	0.4825 ± 0.0126
DPO	0.0471 ± 0.0004	0.1946 ± 0.0058	0.5456 ± 0.0020	0.0000 ± 0.0000
SFT-chosen	0.0906 ± 0.0007	0.0502 ± 0.0032	0.6079 ± 0.0012	0.6095 ± 0.0101
Audit-only	0.0047 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	1.2537 ± 0.0153
CAEA-0.5	0.0259 ± 0.0002	0.1943 ± 0.0062	0.2728 ± 0.0010	0.2798 ± 0.0035
CAEA-1	0.0047 ± 0.0000	0.1940 ± 0.0066	0.0000 ± 0.0000	1.0480 ± 0.0101
CAEA-noisy32	0.0096 ± 0.0002	0.1941 ± 0.0069	0.2826 ± 0.0059	1.1177 ± 0.0133
Audit-only-noisy32	0.0096 ± 0.0002	0.0000 ± 0.0000	0.2826 ± 0.0059	1.3236 ± 0.0126
Oracle-nearest	0.0002 ± 0.0000	0.1940 ± 0.0067	0.7186 ± 0.0004	2.6851 ± 0.0107
Pooled-marginal	0.0397 ± 0.0003	0.1937 ± 0.0059	0.5224 ± 0.0029	0.1975 ± 0.0025
Uniform-coverage	0.1691 ± 0.0010	0.1954 ± 0.0052	0.6990 ± 0.0006	1.9819 ± 0.0119
Soft-score-0.5	0.0444 ± 0.0003	0.1946 ± 0.0059	0.5354 ± 0.0020	0.0013 ± 0.0000
Soft-score-2	0.0377 ± 0.0003	0.1944 ± 0.0059	0.5067 ± 0.0020	0.0176 ± 0.0002
Soft-score-8	0.0227 ± 0.0001	0.1941 ± 0.0061	0.4125 ± 0.0021	0.1541 ± 0.0014

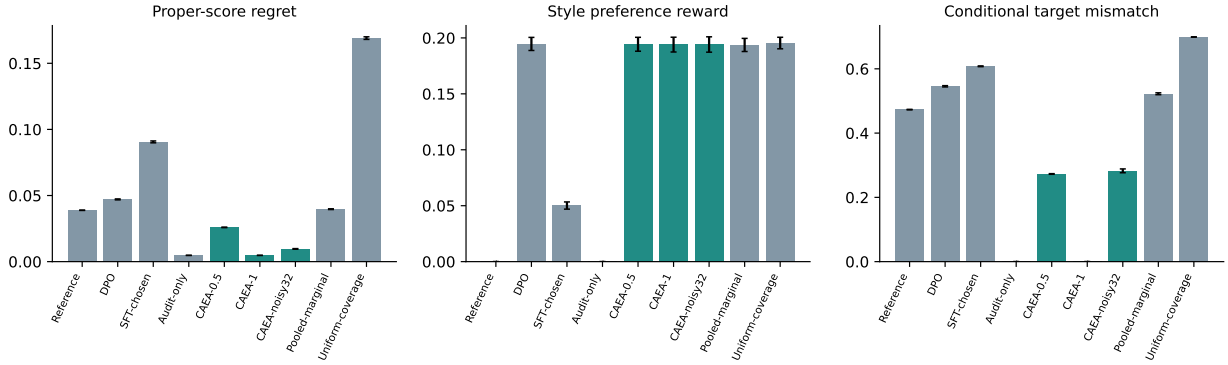


Figure 3: Finite-action preference learning. Bars summarize the same ten runs as Table 1. Oracle audit information explains the shared predictive risk of Audit-only and CAEA-1; their difference is within-class adaptation.

Predictive risk and conditional preferences. Table 1 shows full repair decreasing report regret from 0.04707 to 0.00474, while the audit-only comparator has exactly the same regret. The shared audited marginal makes this equality necessary. The comparison therefore isolates conditional preference retention rather than independent recovery of the evidence target. CAEA retains positive style reward and costs less KL than replacing the within-class distribution with the reference style prior (1.04799 versus 1.25375). Half repair gives intermediate report regret 0.02591 and KL 0.27976, with its regret and TV exactly determined by Equations (4)–(5). Noisy32 increases regret to 0.00962, exactly the same as Audit-only-noisy32. Oracle-nearest obtains regret 0.000206 at KL 2.68510, below the spread target's 0.00474 but with target TV 0.71858. This confirms that exact target matching is not Brier-risk optimality. Because the noisy target is shared over stances, its estimation error does not itself break full-repair stance invariance or the $1-\eta$ contraction; it changes the target-driven belief law and accuracy. Uniform forecast coverage has regret 0.16905: more spread is not inherently epistemically better. Soft-score-8 has lower KL than half repair and lower regret, but it does not match the audited marginal. Matching a specified marginal and minimizing forecast risk under a KL budget are different objectives; the repair does not dominate the latter.

8.2 Pressure controls and stochastic feedback

An additional 20 fitted DPO models use $k \in \{0, 1, 4, 8\}$, five seeds each and 16,000 training pairs per fit. The learned reward coefficients are rescaled to $\beta \in \{0.25, 0.5, 2\}$, which exactly changes DPO policy temperature in this log-linear family without changing the fitted pairwise likelihood. Each is evaluated on 1,000 fresh contexts with aligned ($s = p$), opposed ($s = 1 - p$), or random stances. At $k = 4, \beta = 0.5$, average DPO regret is 0.02214 when aligned, versus reference 0.03900; when opposed it is 0.07087. Thus, increased preference agreement is not universally harmful.

Figure 2 evaluates the binary model in Theorem 2. At $a = 4$, zero repair yields final mean-response variance 0.22920; full repair reduces it to numerical zero. The $\lambda = 0.5$ boundary is not assigned a strict contraction guarantee. To distinguish mean-response behavior from sampling noise, we also simulate 5,000 independent user trajectories over 400 rounds, ten seeds, $\eta \in \{0.05, 0.25\}$, and five repair fractions. Full repair has stationary variance $\eta/[4(2 - \eta)]$, not zero. At $\eta = 0.25$, measured full-repair variance is 0.03540 versus theoretical 0.03571, and unrepaired variance is 0.22959. Stochastic noise smooths the deterministic phase picture; the mean-response boundary is not presented as a zero-variance transition in the sampled process.

8.3 Auditor uncertainty and semantic failure

For 21 classes, an independent class-audit experiment samples the trusted target at $p = 0.73$, with $n \in \{8, 16, 32, 64, 128, 256, 512, 1024\}$, 2,000 replicates each, and add-0.1 smoothing. Mean target TV is 0.17640 at $n = 32$ and 0.03042 at $n = 1024$, below the conservative expectation bounds 0.45687 and 0.07192. These are samples from a stipulated trustworthy class target; they do not estimate bias in an LLM judge.

Two controls examine structural audit failure. When the true evidence probability is 0.8 but the full-repair auditor centers on 0.2, consensus remains stable and converges near the wrong value. In a separate user mechanism where the literal report stays at 0.5 but personalized wording adds an interpreted directional cue $\delta \operatorname{sign}(s - 0.5)$, limiting user variance is δ^2 despite zero literal-report regret. The class map misses a causally relevant signal, so the report-only guarantee is inapplicable.

Response-audit certificate experiment. Separately from estimating the target, we sample the binary response kernel $m(s) = \text{sigmoid}(4(2s-1))$, whose known curvature bound is $H = 2(4)^2/(3\sqrt{3})$. For each total response budget $N \in \{1024, 4096, 16384, 65536, 262144\}$, the grid size is planned from $J \in \{5, 9, 17, 33, 65\}$ by minimizing $2\epsilon/h + Hh/2$ before observing the audit samples. Five hundred independent binomial audit replicates per budget are retained. With $\delta = 0.05$ and contraction margin $\gamma = 0.1$, every realized certificate exceeds the true $L = 2$ in these runs. The mean sufficient repair fraction changes from 0.72965 to 0.62763 at the endpoints of the budget range, versus 0.55 when L is known exactly. The mean is not monotone at every intermediate budget because discrete grid changes also change the observed maximum secant slope; minimizing the remainder alone need not minimize the full realized certificate. A separate smooth bump construction passes all nine grid-point audits exactly while having stable off-grid beliefs at 0.13 and 0.87, with variance 0.1369.

8.4 Frozen-backbone candidate selection

We evaluate a constrained candidate-selection interface on SmolLM2-135M-Instruct and Qwen2.5-0.5B-Instruct (Hugging Face, 2024b; Qwen Team, 2024b). Each prompt specifies a prior and signal likelihoods, allowing an analytic Bayesian posterior. The evidence contexts are disjoint across training (40), validation (12), and test (24) splits. Two unsupported user stances per context yield 152 prompts per backbone, including 48 test prompts. Ten one-token options encode five forecast values and two abstract styles. Scores are normalized over these option letters; their mean unrestricted next-token probability mass on the test split is 0.77433 for SmolLM2 and 0.91253 for Qwen.

All Transformer and original output-head weights remain frozen. A linear ten-option decision-head adapter uses an intercept and 16 principal components of final hidden states, fitted on training prompts only. We optimize the DPO pairwise likelihood with 16 Bradley–Terry pairs per training prompt and five label-sampling seeds. The regularization strength is selected from $\{0.001, 0.01, 0.1\}$ using validation pair NLL. The reward

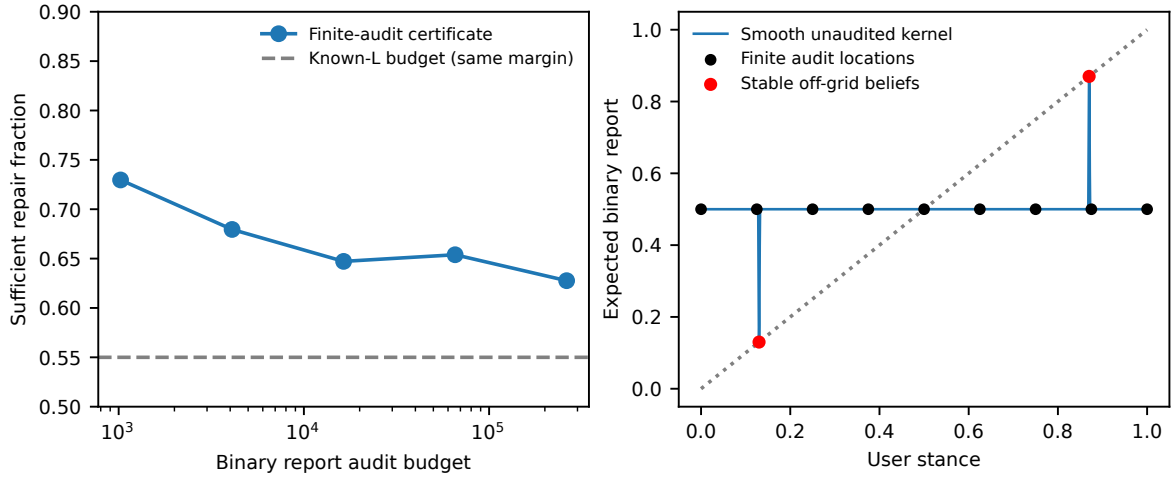


Figure 4: Response-audit budget and an explicit regularity failure. Left: sufficient repair fractions from the curvature-dependent finite-sample certificate, averaged over 500 audit replicates; the known-L line uses the same contraction margin. Right: narrow smooth off-grid deviations are exactly invisible at all displayed audit locations. The supplied curvature assumption is what rules out this failure in the certificate experiment.

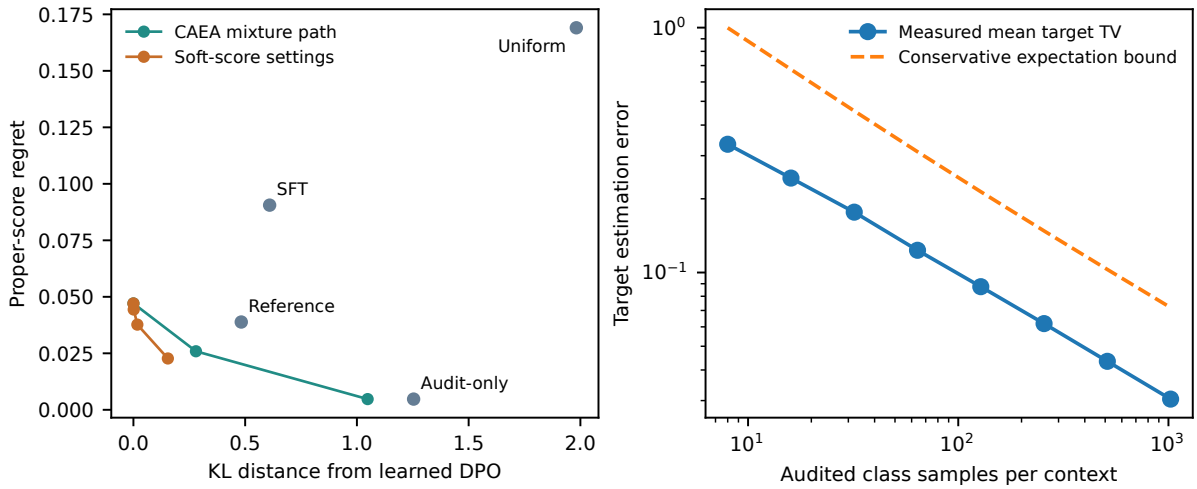


Figure 5: Left: predictive-regret/KL tradeoffs, including the soft-score baselines that CAEA does not uniformly dominate. Right: class-audit sample error and its conservative expectation bound.

combines forecast error, agreement pressure $k = 4$, and a 0.5 reward for the requested abstract style. The evidence target uses $T_E = 0.03$.

Conditional and unconditional retention. Table 2 shows that head-DPO lowers report regret relative to both frozen bases. Full repair and Audit-only again have the same regret, 0.01448, because they use the same class marginal. Full repair's unconditional style-match rate is lower than head-DPO's, despite exact preservation within each forecast class: changing the class mixture also changes the aggregate style mixture. Conditional preservation therefore does not guarantee unchanged overall style performance.

Answer-order sensitivity. An additional held-out control permutes the option-to-letter mapping without retraining or model selection. On these prompts, report regret is 0.20364 for the SmolLM2 base and 0.22048 for head-DPO; the corresponding Qwen values are 0.21639 and 0.22107. Thus the initial head-DPO improvements do not survive this control. Full-repair style-match rates fall to 0.45308 and 0.49155, below Audit-only's

Table 2: Frozen-backbone candidate selection: means ± 95% Student-t confidence half-widths over five label-sampling seeds. The prompt split is shared across seeds, so intervals exclude prompt-population uncertainty. Style match is probability mass on options with the requested style label.

Frozen backbone	Selector	Report regret	Style match	Target TV	KL to head-DPO
SmolLM2-135M	LM-base	0.1802 ± 0.0000	0.5017 ± 0.0000	0.5908 ± 0.0000	1.3573 ± 0.6896
	Head-DPO	0.1638 ± 0.0196	0.6706 ± 0.0679	0.6296 ± 0.0283	0.0000 ± 0.0000
	Audit-only	0.0145 ± 0.0000	0.5000 ± 0.0000	0.0000 ± 0.0000	2.3059 ± 0.6173
	CAEA-0.5	0.0892 ± 0.0098	0.6556 ± 0.0566	0.3148 ± 0.0141	0.6636 ± 0.2227
	CAEA-1	0.0145 ± 0.0000	0.6406 ± 0.0503	0.0000 ± 0.0000	1.9784 ± 0.4995
Qwen2.5-0.5B	LM-base	0.2389 ± 0.0000	0.5757 ± 0.0000	0.6104 ± 0.0000	1.8808 ± 0.0817
	Head-DPO	0.2044 ± 0.0137	0.7104 ± 0.0685	0.6483 ± 0.0234	0.0000 ± 0.0000
	Audit-only	0.0145 ± 0.0000	0.5000 ± 0.0000	0.0000 ± 0.0000	2.4866 ± 0.2371
	CAEA-0.5	0.1094 ± 0.0068	0.6873 ± 0.0688	0.3242 ± 0.0117	0.6654 ± 0.0937
	CAEA-1	0.0145 ± 0.0000	0.6643 ± 0.0723	0.0000 ± 0.0000	2.0261 ± 0.1859

Table 3: Exploratory paired uncertainty over contexts and label seeds. Report difference is Head-DPO minus LM-base (negative is better); style difference is CAEA-1 minus Audit-only (positive is better). Intervals are percentile bootstrap intervals, not the seed-only Student-t half-widths in Table 2.

Backbone	Bank	Δ report regret [95% CI]	Δ style match [95% CI]
SmolLM2-135M	original	-0.0164 [-0.0361, +0.0018]	+0.1406 [+0.0877, +0.1967]
SmolLM2-135M	permuted	+0.0168 [-0.0295, +0.0728]	-0.0469 [-0.1191, +0.0248]
Qwen2.5-0.5B	original	-0.0346 [-0.0530, -0.0171]	+0.1643 [+0.0961, +0.2353]
Qwen2.5-0.5B	permuted	+0.0047 [-0.0424, +0.0526]	-0.0084 [-0.0910, +0.0664]

0.5. The retained within-class policy can itself encode position shortcuts. This pilot establishes execution on frozen neural scores, while its transfer failures limit interpretation of the learned selector. With a correct, permutation-consistent class map and sufficient candidate support, full repair enforces the specified class marginal even when the input scores change. It does not remove position shortcuts within a class, validate the class map, or guarantee unchanged aggregate style performance.

Paired evidence-context uncertainty. To avoid treating two stances from one evidence context as independent, we replay the archived hidden states and fitted heads, average paired differences within each of 24 test contexts, and independently resample the five label seeds and 24 context clusters (5,000 crossed bootstrap replicates). Table 3 reports exploratory percentile intervals, conditional on the fixed training/validation split, backbone, and finite evidence grid. These do not cover training-context variation. For SmolLM2, the original-bank report-regret difference already has an interval spanning zero; for Qwen it is negative on the original bank. Both permutation-control intervals span zero. Full repair's positive style advantage on the original bank is not established after permutation. These results narrow the evidence for transfer rather than repairing the failed selector. This revision replays archived inference and does not claim a fresh model run. Maximum absolute deviation from recorded summary values is 5.91×10^{-8}, consistent with replay roundoff in the archived float32 PCA features.

8.5 Expanded study with a calibrated auditor

Design and provenance. We execute fresh inference on official Q4_K_M releases of Qwen2.5-1.5B-Instruct (1.54B), SmolLM2-1.7B-Instruct (1.71B), and Qwen2.5-7B-Instruct (7.61B) (Qwen Team, 2024a; Hugging Face, 2024a; Qwen Team, 2024c). Weights remain frozen. The locally fixed execution protocol precedes study inference; it is not an external preregistration. Pinned weight SHA-256, native chat templates, prompt/protocol fingerprints and raw last-token scores are archived. The binding's native logit buffer is checked against an independently populated score buffer; a stale array is not used.

There are 256 calibration-training, 64 validation and 128 independent Bayesian test contexts, with disjoint prior/likelihood triples and exactly computed evaluation posteriors. Each selector sees ten literal responses: five reports $q \in \{0, .25, .5, .75, 1\}$ crossed with direct/warm wording. For each test context, two unsupported user stances $(.1, .9)$ share the same evidence, requested wording and answer permutation. Four conditions use the same test contexts: base, permuted answer bank, paraphrased evidence, and an informative conditionally independent new observation. Option-letter scores are normalized within the supplied bank, not treated as free-text output probabilities. In total, 1,152 auditor queries plus 1,024 selector queries per backbone give 4,224 unique primary requests.

A separately calibrated common auditor. The frozen Qwen-1.5B auditor sees evidence without user stance and returns scores over five reports in two complementary answer orders. Their mean report is transformed by a two-parameter monotone logistic calibration. Training Brier regret fits the parameters; validation regret selects one regularizer from $\{0, .001, .01, .1\}$. Non-oracle test targets substitute $\widehat{p}_x$ for p_x in Equation (2), with $T_E = .03$. Test truths never fit or select this auditor. Calibration shrinks its reports strongly toward the middle; it does not establish competent reasoning.

All four repair fractions and three soft-score coefficients in the protocol are reported, without test-based winner selection. Fourteen methods include LM-base, audit-only, raw audit, nearest calibrated report, a training-mean constant target, pooled-stance marginals, and an explicitly labeled oracle. Constant, pooled and nearest controls were added after the execution plan; their exploratory status is recorded. Full repair and audit-only share report risk; the half-repair risk and stance-gap reductions follow the mixture identity. Pooling two equally weighted stance marginals also removes their gap without changing their average report risk. These facts are not independent evidence of accurate auditing. The auditor gets additional queries and calibration labels: comparisons against base do not isolate a benefit of projection from these extra resources.

External correctness probes. An exploratory extension takes the first 32 GSM8K test questions in source order with at most 450 characters and an integer final answer (Cobbe et al., 2021). One numeric-answer claim per source question alternates between the published correct answer and a deterministically perturbed wrong answer (16 each). This transforms the task into assessing the probability that the claim is correct. Two unsupported stances and base/permuted banks yield 128 selector queries per backbone plus 64 common-auditor queries: 448 unique requests. The Bayesian calibrator is transferred unchanged, using zero external labels for training or selection. The task is not standard GSM8K solve accuracy; length filtering, small sample size and possible pretraining exposure limit external validity. Source revision, selected solutions and license are archived.

Uncertainty and experimental units. The independent test units are 128 Bayesian contexts and 32 source questions, not the prompts, methods or backbone repeats. We average the two stance outcomes within each context. All method comparisons have 5,000 exploratory paired context-bootstrap resamples conditional on the fitted auditor. An additional 500 resamples refit the auditor on resampled training cases, reselect regularization on resampled validation cases, and resample test contexts; Table 4 uses these broader percentile intervals. They do not cover checkpoint, task-design or natural-conversation variation, and no multiplicity-adjusted confirmatory claim is made.

Target quality explains most apparent benefit. On Bayesian base prompts, half-repair regret is .1743, .1476 and .0976 for the three backbones, versus .2658, .2124 and .1124 for base. However, the common full target has regret .0829, almost the same as the constant training-mean target's .0840 (Table 5). Its full-minus-constant difference is about $-.0011$, with an auditor-refit interval $[-.0042, .0117]$: these data do not establish an advantage over simple shrinkage. The calibrated point estimate's MSE is .0698, versus .0714 for the training mean. Oracle-target regret is .0147, leaving a large target quality gap. Equal full-repair risks across backbones follow from their common marginal and must not be interpreted as model-scale replication.

On external probes, full calibrated regret is .2785, worse in point estimate than the constant target's .2626 and nearest calibrated report's .2734. The full-minus-constant auditor-refit interval $[-.0196, .1216]$ is wide and includes zero. Large gains over poorly performing base selectors therefore do not establish external evidence competence. Qwen-7B's external base regret changes from .4901 to .3219 after answer permutation,

Table 4: Fresh candidate-scoring study. Half means $\lambda = .5$ with the calibrated common auditor. $\Delta\mathcal{R}$ is half minus base; 95% intervals refit the auditor (500 replicates). Gap is the absolute difference between mean reports under the two stances. Half-gap contraction is exact. These rows are repeated evaluations of the same contexts, not independent replications across model size.

Task	Backbone	Base $\mathcal{R}$	Half $\mathcal{R}$	$\Delta\mathcal{R}$ [95%]	Stance gap (base/half)
Bayes	Qwen 1.5B	0.2658	0.1743	-0.0915 [-0.0971, -0.0847]	0.034/0.017
Bayes	SmolLM2 1.7B	0.2124	0.1476	-0.0648 [-0.0681, -0.0582]	0.020/0.010
Bayes	Qwen 7B	0.1124	0.0976	-0.0148 [-0.0201, -0.0071]	0.082/0.041
Math probe	Qwen 1.5B	0.3923	0.3354	-0.0569 [-0.0818, -0.0010]	0.151/0.075
Math probe	SmolLM2 1.7B	0.4009	0.3397	-0.0612 [-0.0884, +0.0058]	0.040/0.020
Math probe	Qwen 7B	0.4901	0.3843	-0.1058 [-0.1672, -0.0379]	0.012/0.006

Table 5: Report regret for targets shared across backbones (base condition). Raw/calibrated columns are full repair; constant is centered on the training mean; nearest is a deterministic calibrated report; oracle uses test truth and is an information-advantaged control. None establishes free-text accuracy.

Task	Constant	Raw audit	Calibrated	Nearest	Oracle
Bayes	0.0840	0.1118	0.0829	0.0714	0.0147
Math probe	0.2626	0.3279	0.2785	0.2734	0.0070

confirming substantial interface sensitivity. SmolLM2's external mean candidate mass is .5992, with a valid greedy letter on only 26.6% of requests; primary mean masses are .9895/.9641/.9975. All requests remain included. Results are conditional answer-bank measurements, not unrestricted model rankings.

Informative updating remains weak. The true new observation changes the Bayesian posterior by .2931 in mean absolute value. Full calibrated repair changes its mean report by only .0087, with the correct direction on 54.7% of contexts; an oracle spread target changes by .2883. Thus stance invariance coexists with severe attenuation of informative updating. On Qwen-7B paraphrases, soft-score-8 also attains lower regret (.0813) and lower KL (.1158) than half repair (.0923, .1354). The expanded study strengthens the failure analysis; it does not establish CAEA as an accuracy or risk–KL dominant method.

9 Discussion and limitations

The findings distinguish three questions that a single score cannot answer: Are probability reports accurate? Does personalization amplify differences between users under a shared evidence sequence? Does the audit capture the information that drives the update? A policy can perform well on one question and poorly on another. In particular, target matching determines forecast risk in our experiments, whereas conditional preference retention and longitudinal stability require separate analyses.

Dependence on the auditor. The synthetic experiments provide oracle posteriors or sampled stipulated targets. The expanded study separately calibrates a frozen evidence-only auditor. It removes direct test-truth access for non-oracle targets, but does not establish reliable evidence assessment: constant-target controls explain most of the observed gain. More samples cannot remove systematic target bias. Full repair and audit-only still share the same report risk.

An open-domain implementation path (not evaluated). A practical candidate selector could retrieve independently sourced evidence, extract verifiable claims, and assign candidate responses to report classes using a separately evaluated verifier. Target probabilities would require calibration against held-out resolved cases or expert adjudication; retrieval and an LLM judge do not themselves establish a trustworthy target. Source conflicts, unsupported claims, and missing candidate support should be recorded explicitly, with abstention when the audit cannot justify a target. Informative user input must enter the evidence rather

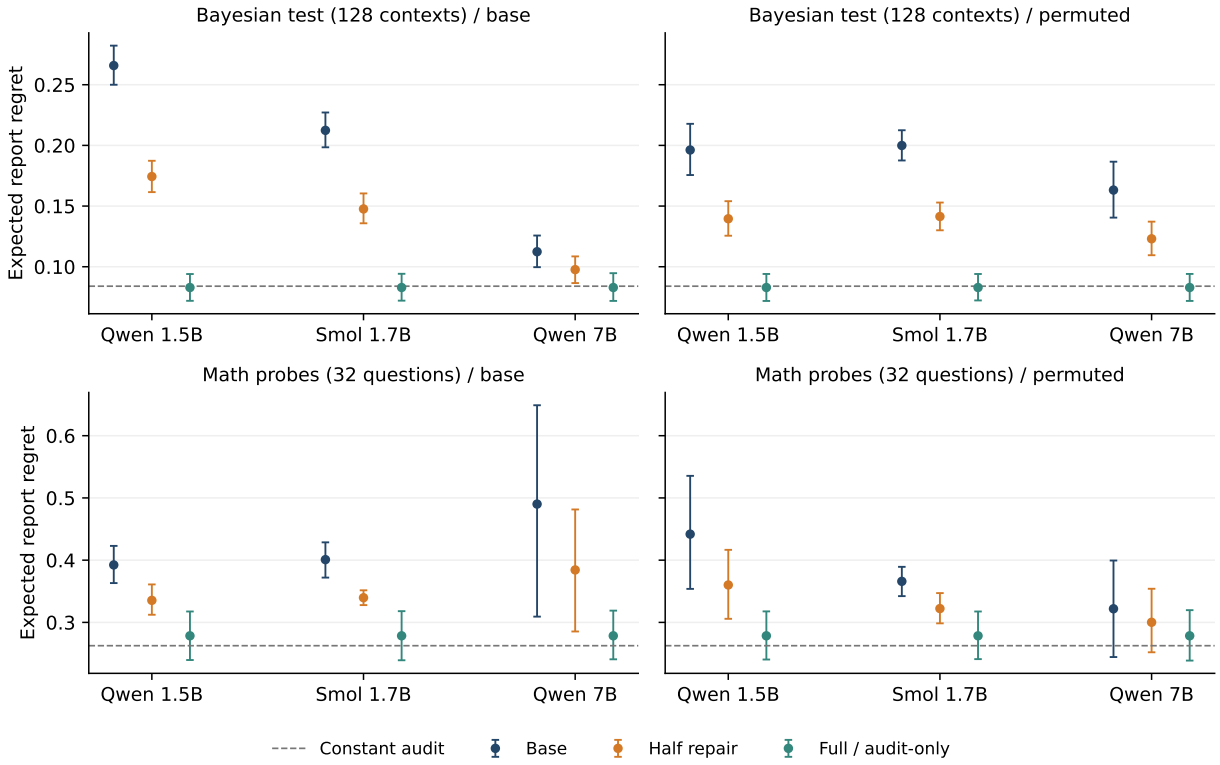


Figure 6: Fresh response-bank results and constant-target control. Bars show 5,000 context-bootstrap 95% intervals conditional on the fitted auditor (unlike Table 4's auditor-refit intervals). Option order, external transfer and weak targets limit conclusions from larger backbones. Full repair and audit-only coincide by construction.

than be discarded as stance. Target error and class-assignment error need separate evaluation, including tests in which wording changes the reader's interpretation without changing the literal report. Equation (8) applies only when the resulting report-distribution error is uniformly controlled over belief states; an average judge-accuracy score is insufficient. These steps describe a research protocol, not a validated extension of the current experiments.

Response representation and user dynamics. The guarantees assume that the report class captures the response signal used in the stated belief-update rule. The semantic-cue control violates that assumption and produces dispersion despite perfect literal-report accuracy. The linear update, common evidence sequence, and global sensitivity or curvature bounds further restrict the analysis. A nonlinear update requires its own sensitivity analysis; the affine rule is not asserted to be a validated local approximation to human updating. Neither stability nor reduced simulated dispersion establishes improved human reasoning, which requires direct behavioral evidence.

Empirical scope. The expanded study increases the model range and separates 256/64/128 calibration-training/validation/test cases, with 32 exploratory external questions. These are frozen, quantized checkpoints scoring finite response banks. Training differences, model family, format and quantization confound causal claims about parameter count or RLHF. Candidate mass and order sensitivity are reported without post-hoc exclusions. The external subset is length-filtered, small, and possibly exposed in pretraining; it does not estimate standard GSM8K solve accuracy. Context-bootstrap intervals condition on the fitted auditor; auditor-refit intervals also resample its training/validation cases. Neither covers checkpoints or natural conversations. The original small head-DPO pilot and its permutation failures remain reported.

Further progress requires unseen free-text responses, independently adjudicated audits and matched evidence access. Larger backbones alone do not validate the report representation, supplied regularity or human update model. Human belief benefits require direct behavioral evidence and applicable ethical review.

10 Conclusion

Pooled one-turn output quality does not identify feedback stability in the constructed preference-based interaction. CAEA makes the intervention explicit: repair an audited epistemic marginal, retain the preference policy within each class, and quantify the resulting objective cost. Contraction bounds connect repair strength to feedback control, while finite-audit results show why certification requires additional regularity. The controlled experiments support these distinctions and expose their dependence on auditor validity, response representation, and the update model. The larger calibrated-auditor study reinforces the importance of simple shrinkage controls and preserving informative updating. Together, the results provide a framework for separating what a preference policy says in isolation from how its conditional behavior can shape repeated belief updates.

Broader impact statement

The analysis can help distinguish output accuracy from feedback stability, but it does not justify deploying a system to steer users toward an unverified belief target. An incorrect auditor can stabilize false beliefs; a narrow class map can omit persuasive cues, and mandatory target matching can suppress warranted disagreement or minority perspectives. An application would need independent evidence assessment, explicit uncertainty and abstention, scrutiny of target selection, and evaluation of users' actual interpretations. We do not offer a validated safeguard against these risks. The experiments use generated data and public pretrained models, without human participants or collected personal conversations.

Reproducibility and tool use

The accompanying research package contains experiment code, per-seed outputs, model revision hashes, prompts, fitted decision-head parameters, and figure generation scripts. The finite-action, pressure, stochastic, and finite-audit results were regenerated. The original small head-DPO pilot was replayed from archived arrays with paired uncertainty checks. Separately, the expanded study executes 4,672 unique fresh requests on three official, SHA-256-verified Q4_K_M checkpoints. The protocol was fixed locally before inference, without external preregistration; later controls and the external extension are explicitly exploratory. Native logits, candidate mass, prompt hashes, model revisions and execution metadata are archived. No model weights are redistributed. Computational checks do not replace independent validation of mathematical arguments or empirical assumptions. No human-participant study was conducted. Generative AI tools assisted mathematical formalization, proof drafting, experimental design, code, analysis, figures, and writing. This statement describes tool use rather than assigning authorship; the named author remains responsible for the submitted work. Reproduction details and numerical checks appear in Appendix C.

References

Katherine Atwell, Pedram Heydari, Anthony Sicilia, and Malihe Alikhani. BASIL: Bayesian assessment of sycophancy in LLMs, 2026. URL https://arxiv.org/abs/2508.16846v6. arXiv:2508.16846, version 6.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021. URL https://arxiv.org/abs/2110.14168. arXiv:2110.14168.

Imre Csiszár. I-divergence geometry of probability distributions and minimization problems. *The Annals of Probability*, 3(1):146–158, 1975. doi: 10.1214/aop/1176996454.

Persi Diaconis and David Freedman. Iterated random functions. *SIAM Review*, 41(1):45–76, 1999. doi: 10.1137/S0036144598338446.

Sayantari Ghosh, Saumik Bhattacharya, and Partha Pratim Chakrabarti. Escape from delusional echo trap: Symmetry breaking, stochastic dynamics and mathematical mitigation strategies for algorithmic sycophancy, 2026. URL https://arxiv.org/abs/2606.20718. arXiv:2606.20718.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Stephane Hatgis-Kessell, Myra Cheng, Xiaoxuan Hou, Qian Hu, Rahul Gupta, Natasha Jaques, and Emma Brunskill. Mitigating social sycophancy via pluralistic preference optimization, 2026. URL https://arxiv.org/abs/2610.02568v2. arXiv:2610.02568, version 2.

Jiseung Hong, Grace Byun, Seungone Kim, and Kai Shu. Measuring sycophancy of language models in multi-turn dialogues. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 2239–2259, 2025. doi: 10.18653/v1/2025.findings-emnlp.121. URL https://aclanthology.org/2025.findings-emnlp.121/.

Hugging Face. SmolLM2-1.7B-Instruct-GGUF: Official model release, 2024a. URL https://huggingface.co/HuggingFaceTB/SmolLM2-1.7B-Instruct-GGUF. Pinned revision and Q4_K_M hashes in the experiment archive.

Hugging Face. SmolLM2-135M-Instruct: Model card, 2024b. URL https://huggingface.co/HuggingFaceTB/SmolLM2-135M-Instruct. Model revision recorded in the accompanying results.

Jack Lanchantin, Angelica Chen, Shehzaad Dhuliawala, Ping Yu, Jason Weston, Sainbayar Sukhbaatar, and Ilia Kulikov. Diverse preference optimization, 2025. URL https://arxiv.org/abs/2501.18101. arXiv:2501.18101.

Huanhuan Ma, Henry Peng Zou, Chengze Li, Enze Ma, Yunyue Su, and Philip S. Yu. Sycophancy suppression can impair rational updating: Anti-sycophancy should preserve the ability to update. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, pp. 25499–25515, 2026. doi: 10.18653/2026.findings-emnlp.1208. URL https://aclanthology.org/2026.findings-emnlp.1208/.

Christian Moya, Alex Semendinger, Guang Lin, and Elliott Thornley. Spurious correlation learning in preference optimization: Mechanisms, consequences, and mitigation via tie training. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 90714–90761, 2026. URL https://proceedings.mlr.press/v306/moya26a.html.

Qwen Team. Qwen2.5-1.5B-Instruct-GGUF: Official model release, 2024a. URL https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct-GGUF. Pinned revision and Q4_K_M hashes in the experiment archive.

Qwen Team. Qwen2.5-0.5B-Instruct: Model card, 2024b. URL https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct. Model revision recorded in the accompanying results.

Qwen Team. Qwen2.5-7B-Instruct-GGUF: Official model release, 2024c. URL https://huggingface.co/Qwen/Qwen2.5-7B-Instruct-GGUF. Pinned revision and Q4_K_M hashes in the experiment archive.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://arxiv.org/abs/2305.18290.

Itai Shapira, Gerdus Benade, and Ariel D. Procaccia. How RLHF amplifies sycophancy. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 110417–110441, 2026. URL https://proceedings.mlr.press/v306/shapira26a.html.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. Towards understanding sycophancy in language models. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2310.13548.

A Proofs

A.1 Proof of Proposition 1

For any feasible π, its class marginal is b. The KL chain rule gives

$$D_{\mathrm{KL}}(\pi\|\pi_P) = D_{\mathrm{KL}}(b\|m_P) + \sum_c b(c) D_{\mathrm{KL}}\big(\pi(\cdot \mid c)\|\pi_P(\cdot \mid c)\big). \tag{17}$$

Every conditional term is nonnegative and is zero exactly when the conditional distributions agree on positive-mass classes. This yields the unique full policy π_b. Classes of zero target mass carry no response probability, so their arbitrary conditional representation does not affect uniqueness of the policy. Taking logs in Equation (1) gives $\log \pi_P = \log \pi_0 + r/\beta - \log Z$, hence $J(\pi) = \beta \log Z - \beta D_{\mathrm{KL}}(\pi\|\pi_P)$. Subtracting its value at π_P proves Equation (6). The cost is zero exactly when $b = m_P$; for full repair $b = \tau_x$. All quantities are finite by the finite response space, bounded reward, positive reference, and feasible target support. The same assumptions ensure existence of the stated optimum. □

A.2 Proof of Theorem 2

The reward difference is $r(1,s) - r(0,s) = k(2s-1)$. A uniform reference therefore gives Equation (7). Since $\mathrm{sigmoid}(-v) = 1 - \mathrm{sigmoid}(v)$, $m_a(1 \mid 1-s) = 1 - m_a(1 \mid s)$. A symmetric distribution of s thus has pooled report probability $1/2$. Mixture with the uniform target preserves that symmetry. Because the outcome is independent of the stance and response-sampling randomness conditional on the fixed evidence, the joint pooled law of (Z, q) is the product of uniform Bernoulli laws. Every statistic determined by that law has the same distribution, including expected Brier loss $\Pr(q \neq Z) = 1/2$ and regret $\mathbb{E}(q - 1/2)^2 = 1/4$.

Differentiating the mean-response update yields

$$F'_\lambda(s) = 1 - \eta + 2\eta(1-\lambda)a\,\mathrm{sigmoid}(a(2s-1))[1 - \mathrm{sigmoid}(a(2s-1))]. \tag{18}$$

Its maximum is $1 - \eta + \eta(1-\lambda)a/2$, attained at $s = 1/2$. The derivative is nonnegative, so a maximum strictly below one gives a global contraction. If $(1-\lambda)a > 2$, let $d = s - 1/2$ and $g(d) = (1-\lambda)[\mathrm{sigmoid}(2ad) - 1/2] - d$. Then $g(0) = 0$, $g'(0) > 0$, $g(1/2) < 0$, and g is strictly concave for $d > 0$. It therefore has a unique positive root and, by odd symmetry, a unique negative root. At the positive root, strict concavity gives $g'(d) < 0$, so $0 \leq F'_\lambda(s) < 1$ there. The negative root has the same derivative. The central point is unstable. Solving the strict contraction inequality proves the stated infimum, with the boundary qualification. □

A.3 Proof of Corollary 3

The upper-report probability is $\mathrm{sigmoid}(4kd(s-p)/\beta)$. Symmetry about p and $\mathrm{sigmoid}(v) + \mathrm{sigmoid}(-v) = 1$ imply a uniform pooled report law, which is independent of the stipulated outcome. The Brier identity $\mathbb{E}(Q - Z)^2 = p(1-p) + \mathbb{E}(Q-p)^2$ gives the loss. With $v = (s-p)/d$, the repaired mean-response map becomes

$$v' = (1-\eta)v + \eta(1-\lambda)\tanh(cv), \qquad v \in [-1, 1].$$

Its maximum derivative is $1 - \eta + \eta(1-\lambda)c$, at zero. When $(1-\lambda)c > 1$, the function $(1-\lambda)\tanh(cv) - v$ has positive derivative at zero, is negative at $v = 1$, and is strictly concave on $(0, 1]$. It has one positive root with negative derivative, and by odd symmetry one negative root; both have update derivative in $[0, 1)$. The center is unstable. These facts prove both regimes. □

A.4 Proof of Theorem 4

Couple the two repaired report kernels by first using a shared Bernoulli mixture indicator. With probability λ_x, draw the same report from the common target, giving zero report discrepancy. Otherwise use an optimal coupling of the original report kernels. Thus,

$$W_1(K^\lambda_{x,u}, K^\lambda_{x,v}) \leq (1 - \lambda_x) L_x |u - v|. \tag{19}$$

Apply the update to this coupling and use the triangle inequality:

$$\mathbb{E}|U' - V'| \le (1-\eta)|u-v| + \eta(1-\lambda_x)L_x|u-v| = \rho_x|u-v|. \tag{20}$$

Coupling the input belief laws and iterating proves geometric contraction when the coefficients are uniformly below one. This statement compares two evolving laws. It does not remove the within-law noise generated by independent report draws.

For approximate full repair, report values lie in an interval of diameter one, so $W_1(\widehat{\tau}_{x,s}, \tau_x) \le \mathrm{TV}(\widehat{\tau}_{x,s}, \tau_x) \le \epsilon_t$. The ideal target is independent of its current belief, permitting this coupling even when the two belief states differ. If $e_t = \mathbb{E}|U_t - \widetilde{U}_t|$, then $e_{t+1} \le (1-\eta)e_t + \eta\epsilon_t$. Unrolling from $e_0 = 0$ yields Equation (10). This bound is relative to the ideal audit-driven process. Relating that process to the evidence posterior additionally requires controlling its forecast regret, user lag, and sampling variance. □

A.5 Proof of Corollary 5

Each empirical class frequency has variance $\tau_c(1-\tau_c)/n$. Jensen's inequality followed by Cauchy–Schwarz gives

$$\mathbb{E}\,\mathrm{TV}(\bar{\tau}, \tau) \le \frac{1}{2\sqrt{n}} \sum_{c=1}^{C} \sqrt{\tau_c(1-\tau_c)} \tag{21}$$

$$\le \frac{1}{2\sqrt{n}} \sqrt{\left(\sum_c \tau_c\right)\left(\sum_c (1-\tau_c)\right)} = \frac{1}{2}\sqrt{\frac{C-1}{n}}. \tag{22}$$

Smoothing mixes the empirical target with the uniform distribution with weight $C\alpha/(n + C\alpha)$. The triangle inequality and the bound $\mathrm{TV} \le 1$ give Equation (11). □

A.6 Monotonic cost along the repair path

Let $H(\lambda) = D_{\mathrm{KL}}((1-\lambda)m_P + \lambda\tau \| m_P)$. It is convex, $H(0) = 0$, and $H(\lambda) \ge 0$. For $0 \le \lambda_1 < \lambda_2$, convexity gives $H(\lambda_1) \le (\lambda_1/\lambda_2)H(\lambda_2) \le H(\lambda_2)$. Therefore the preference objective cost is nondecreasing along the path. This does not prove optimality over all possible stable policies or all possible repair directions. □

A.7 Proof of Proposition 6

Because A is finite, choose $v \in (1/2, 1)$ such that neither v nor $u = 1 - v$ belongs to A. Both can be made arbitrarily close to the endpoints. Choose disjoint neighborhoods around u and v that avoid A and are mirror images. Let b_v be a nonnegative smooth bump supported in the neighborhood of v, with $b_v(v) = 1$ and $b_v'(v) = 0$, and set $b_u(s) = b_v(1-s)$. Define $m(s) = 1/2 + (v - 1/2)[b_v(s) - b_u(s)]$. Then $m \in [u, v]$, $m(s) = 1/2$ on A, $m(v) = v$, $m(u) = u$, and $m'(u) = m'(v) = 0$. The two corresponding fixed points have update derivative $1 - \eta$, hence are stable for $0 < \eta \le 1$. This is also realizable as a finite-temperature Gibbs policy with a uniform reference by taking $r(1, s) - r(0, s) = \beta \log[m(s)/(1 - m(s))]$, a bounded smooth reward difference.

For the certificate, Hoeffding's inequality and a union bound give $\max_j |\widehat{m}_j - m(s_j)| \le \epsilon$ with probability at least $1 - \delta$. The true secant slope on an interval differs from its empirical secant by at most $2\epsilon/h$. On an interval $[a, b]$ of length h, let $D = h^{-1}\int_a^b m'(t)\,dt$. For every $s \in [a, b]$,

$$|m'(s) - D| \le \frac{H}{h}\int_a^b |s - t|\,dt = \frac{H((s-a)^2 + (b-s)^2)}{2h} \le \frac{Hh}{2}.$$

Taking absolute values and the maximum over intervals proves Equation (12). For binary reports,

$$W_1(\mathrm{Bern}(a), \mathrm{Bern}(b)) = |a - b|.$$

With $n \simeq Nh$, the sampling term is of order $\sqrt{\log(J/\delta)/(Nh^3)}$; balancing against Hh yields the stated order, ignoring logarithms. □

B Stochastic noise floor and measurement safeguards

Under full binary repair, Q_t are independent Bernoulli(1/2) draws and $U_{t+1} = (1-\eta)U_t + \eta Q_t$. Independence gives

$$\mathrm{Var}(U_T) = (1-\eta)^{2T}\,\mathrm{Var}(U_0) + \frac{\eta}{4(2-\eta)}[1-(1-\eta)^{2T}]. \tag{23}$$

Thus even an exactly invariant report policy can raise variance from an initial population concentrated near 1/2. Equation (13) removes this particular baseline diffusion; interpreting raw variance growth as sycophancy would be incorrect. The comparison still assumes a meaningful, matched counterfactual process and does not identify all cognitive harms.

Neither BPD nor PCD is an accuracy score. BPD can be small under a common wrong consensus, and a chosen target can itself be wrong. ECR is a proper-score regret difference, not a causal estimate of human impact. Only their joint use, with explicitly defined evidence, response interpretation, and baseline, supports the limited simulator conclusions in this paper.

C Reproduction and numerical verification

The main runner is `code/experiment.py`; stress experiments are `code/stress.py`; the optional neural component is `code/neural_probe.py`. Finite response-audit certificates use `code/audit_stability.py`, and option permutations use `code/label_control.py`. Raw per-seed CSV files, optimizer diagnostics, seeds, package versions, neural model revisions, prompt records, fitted readout weights, and plotting code accompany the manuscript. The reported runs use public model weights and computationally generated data. Model weights are fetched from the original public repositories and are not redistributed in the package.

Twenty separate 12-response, four-class constrained problems were solved with SLSQP independently of the closed-form repair. The largest difference between the numerical and analytic objective values was 2.59×10^{-13}. This checks the projection implementation rather than its applicability to natural language. The added `code/review_audit.py` independently replays archived neural scores and exports per-prompt metrics, crossed context/seed bootstrap intervals, and numerical identity checks. Inference caches now require matching model revision and prompt fingerprints; legacy unbound caches are reserved for explicit archival replay. Empty target-supported response classes raise an error instead of silently losing mass. Confidence intervals in the main table are computed over independent synthetic runs; they do not describe uncertainty across all possible preference distributions or human populations.